MONTH 1: BASIC IMPLEMENTATION AND UNDERSTANDING

- Week 1: Data Collection and Preprocessing
  - Project: Collect a dataset from an online source (e.g., Kaggle) and preprocess it using basic NLP techniques. Document the preprocessing steps.

- Week 2: Model Building and Evaluation
  - Project: Implement a basic fake news detection model using Logistic Regression. Compare it with a Decision Tree model and present the results in a report.

- Week 3: Understanding Sentiment Analysis
  - Project: Build a simple sentiment analysis tool that classifies product reviews as positive, negative, or neutral. Visualize the sentiment distribution for a set of reviews.

- Week 4: Model Training and Visualization
  - Project: Train an SVM model for sentiment analysis and compare its performance with a Naive Bayes model. Create a dashboard to visualize the sentiment analysis results.

MONTH 2: ADVANCED TECHNIQUES AND APPLICATIONS

- Week 5: Data Exploration and Feature Engineering
  - Project: Perform feature engineering on a Parkinson's Disease dataset and create a comprehensive feature set. Document the importance of each feature.

- Week 6: Model Development and Evaluation
  - Project: Develop and evaluate a Random Forest classifier for detecting Parkinson's Disease. Compare it with an SVM model and present the findings in a detailed report.

- Week 7: Sequence Modeling and RNN Implementation
  - Project: Implement an RNN-based model to predict the next word in a given sentence. Train it on a simple text dataset and evaluate its performance.

- Week 8: Advanced Model Training
  - Project: Fine-tune a Transformer-based model for next-word prediction. Compare it with the RNN model and analyze the results in a project report.

MONTH 3: INTEGRATION, OPTIMIZATION, AND DEPLOYMENT

Week 9: Optimization Techniques

- Project: Implement a hyperparameter tuning technique (e.g., grid search) to optimize one of the models developed earlier. Document the improvements achieved through optimization.

Week 10: Real-World Application

- Project: Integrate one of the models into a real-world application (e.g., a web app). Create a user interface that interacts with the model, allowing users to input data and see predictions.

Week 11: Deployment of Models

- Project: Deploy the integrated model as a REST API using a cloud service or containerization tool like Docker. Provide a demo of the deployed application.

Week 12: Documentation and Final Project

- Project: Prepare a final comprehensive project that includes all the models, their optimizations, real-world applications, and deployments. Present this project in a well-documented report and demo it to the team.